

Initiating Search

November 4, 2025, 9:57 PM

• Search:

CAS Lexicon Search:

Curtius reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,184)	 References	View Results
Exported: Retrieved Related Reaction Results (978)	 Reactions	View Results

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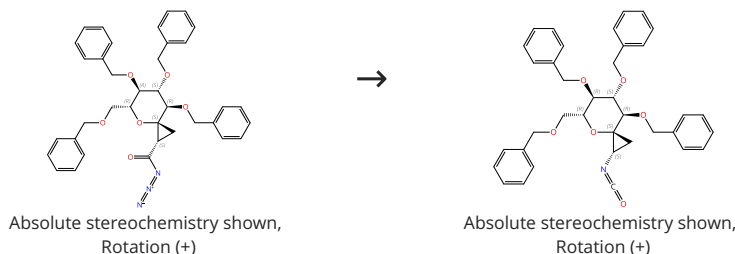
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-7805562

Steps: 1 Yield: 100%

Synthesis of monosaccharide-derived spirocyclic cyclopropylamines and their evaluation as glycosidase inhibitors

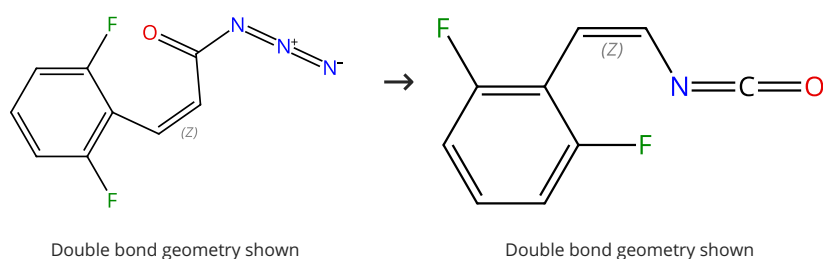
By: Bluechel, Christian; et al

Helvetica Chimica Acta (2003), 86(9), 2998-3036.

1.1 Solvents: Toluene; 2 h, 100 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-6620727

Steps: 1 Yield: 100%

Synthesis of Z-N-alkenylformamides

By: Sanchez, Lucina G.; et al

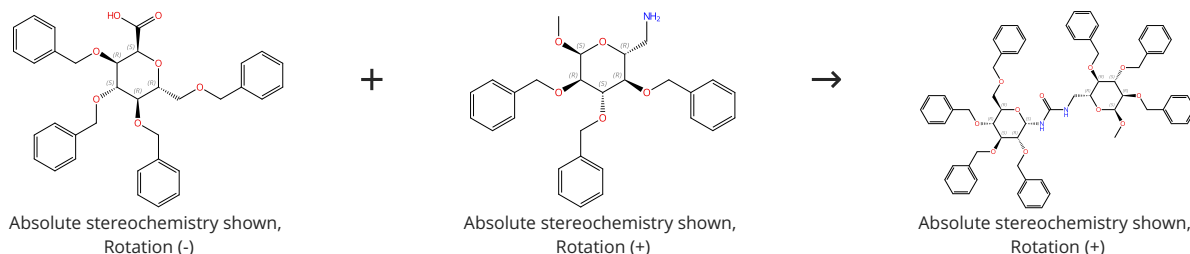
Synthetic Communications (2008), 38(1), 45-53.

1.1 Solvents: Toluene; reflux

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-316-CAS-18081616

Steps: 1 Yield: 100%

Novel synthesis of oligosaccharides linked with carbamate and urea bonds utilizing modified Curtius rearrangement

By: Sawada, Daisuke; et al

Tetrahedron (2008), 64(37), 8780-8788.

1.1 Reagents: Triethylamine, Diphenylphosphoryl azide
Solvents: Benzene; 1 h, reflux

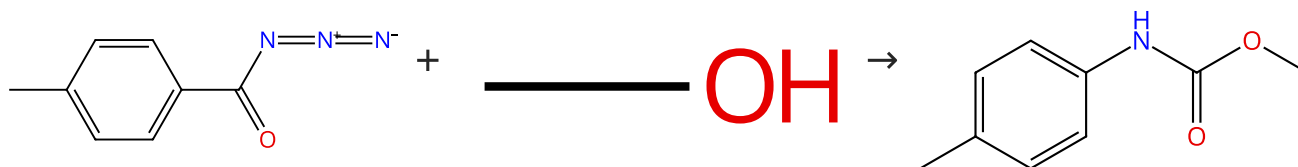
1.2 Solvents: Acetonitrile; rt; 5 h, reflux; reflux → 0 °C

1.3 Reagents: Ammonium chloride
Solvents: Water; 0 °C

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

Suppliers (444)

Suppliers (21)

31-449-CAS-22751546

Steps: 1 Yield: 100%

Synthesis of N-methylated amines from acyl azides using methanol

1.1 Solvents: Methanol; 3 h, 70 °C

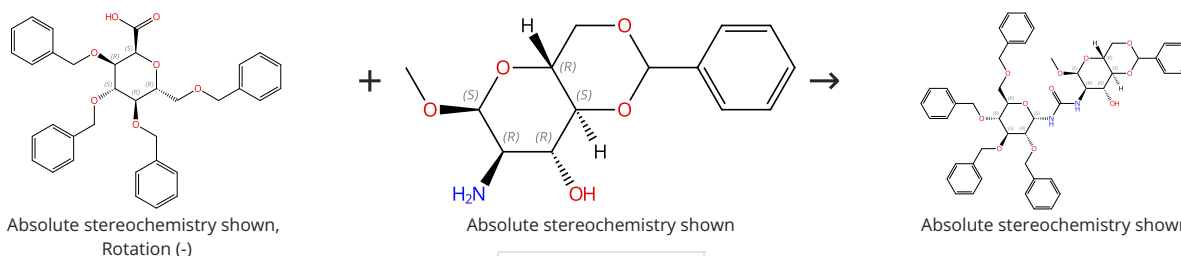
By: Chakrabarti, Kaushik; et al

Experimental Protocols

Organic & Biomolecular Chemistry (2020), 18(30), 5891-5896.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (3)

31-316-CAS-18079833

Steps: 1 Yield: 100%

Novel synthesis of oligosaccharides linked with carbamate and urea bonds utilizing modified Curtius rearrangement

1.1 Reagents: Triethylamine, Diphenylphosphoryl azide

By: Sawada, Daisuke; et al

Solvents: Benzene; 1 h, reflux

1.2 Solvents: Acetonitrile; rt; 5 h, reflux; reflux → 0 °C

Tetrahedron (2008), 64(37), 8780-8788.

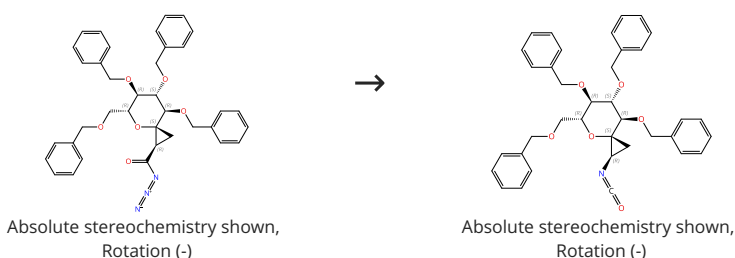
1.3 Reagents: Ammonium chloride

Solvents: Water; 0 °C

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

31-449-CAS-7622616

Steps: 1 Yield: 100%

Synthesis of monosaccharide-derived spirocyclic cyclopropylamines and their evaluation as glycosidase inhibitors

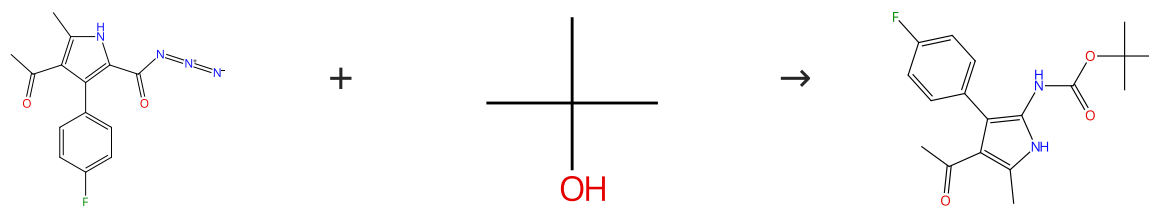
1.1 Solvents: Toluene; 2 h, 100 °C

By: Bluechel, Christian; et al

Helvetica Chimica Acta (2003), 86(9), 2998-3036.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

31-449-CAS-21706391

Steps: 1 Yield: 100%

2H-Azirine-2-carbonyl Azides: Preparation and Use as N-Heterocyclic Building Blocks1.1 Solvents: *tert*-Butanol; 1 h, reflux

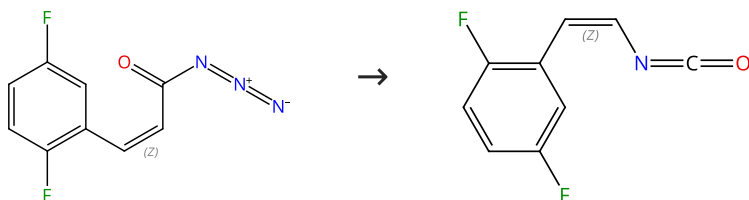
Experimental Protocols

By: Funt, Liya D.; et al

Journal of Organic Chemistry (2020), 85(6), 4182-4194.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

31-449-CAS-4493361

Steps: 1 Yield: 100%

Synthesis of Z-N-alkenylformamides

1.1 Solvents: Toluene; reflux

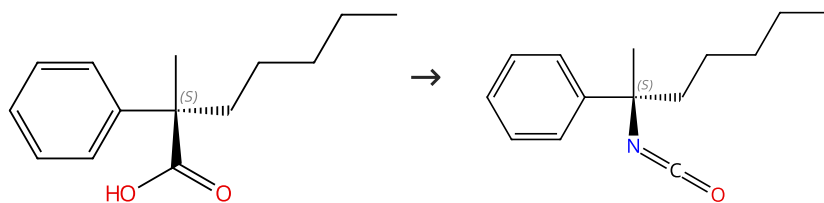
Experimental Protocols

By: Sanchez, Lucina G.; et al

Synthetic Communications (2008), 38(1), 45-53.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

Supplier (1)

31-449-CAS-18101515

Steps: 1 Yield: 100%

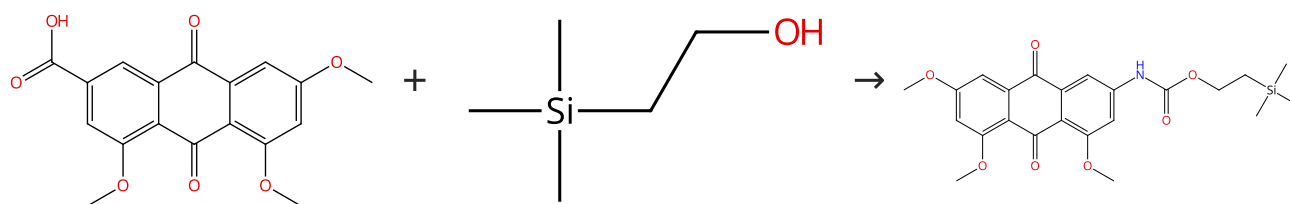
Highly enantioselective preparation of tertiary alcohols and amines by copper-mediated diastereoselective allylic S_N2' substitutions1.1 Reagents: Triethylamine, Diphenylphosphoryl azide
Solvents: Toluene; 2 h, reflux

By: Leuser, Helena; et al

Angewandte Chemie, International Edition (2005), 44(29), 4627-4631.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (85)

31-449-CAS-18042883

Steps: 1 Yield: 100%

An Efficient Route to Emodic Amine and Analogous O-Methyl Protected Derivatives Starting from Emodin

By: Lackner, Bernd; et al

Monatshefte fuer Chemie (2005), 136(9), 1629-1639.

1.1 Reagents: Triethylamine

Solvents: 1,4-Dioxane; 15 min, rt

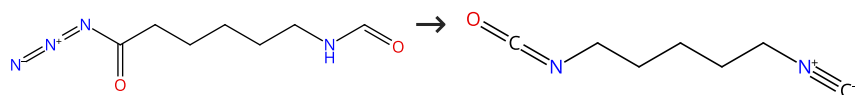
1.2 Reagents: Diphenylphosphoryl azide; 2.5 h, rt

1.3 12 h, 80 °C

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (2)

31-428-CAS-1515743

Steps: 1 Yield: 100%

Preparation of bifunctional isocyanato-isocyanide coupling agents via the phosgenation and dehydration of aminoformamides and related compounds

Assignee: Eckert GmbH

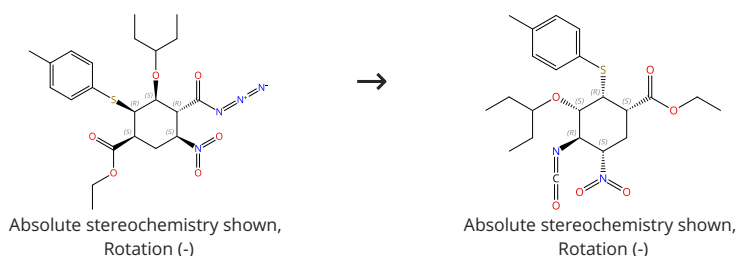
European Patent Organization, EP1300393 A1 2003-04-09

PatentPak available

No Data Available

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

31-449-CAS-13881647

Steps: 1 Yield: 100%

High-Yielding Synthesis of the Anti-Influenza Neuraminidase Inhibitor (-)-Oseltamivir by Two "One-Pot" Sequences

By: Ishikawa, Hayato; et al

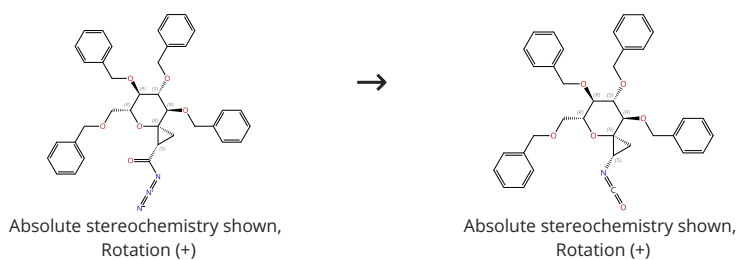
Chemistry - A European Journal (2010), 16(42), 12616-12626, S12616/1-S12616/52.

1.1 Solvents: Benzene-*d*₆; 36 h, 23 °C

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-9932316

Steps: 1 Yield: 100%

Synthesis of monosaccharide-derived spirocyclic cyclopropylamines and their evaluation as glycosidase inhibitors

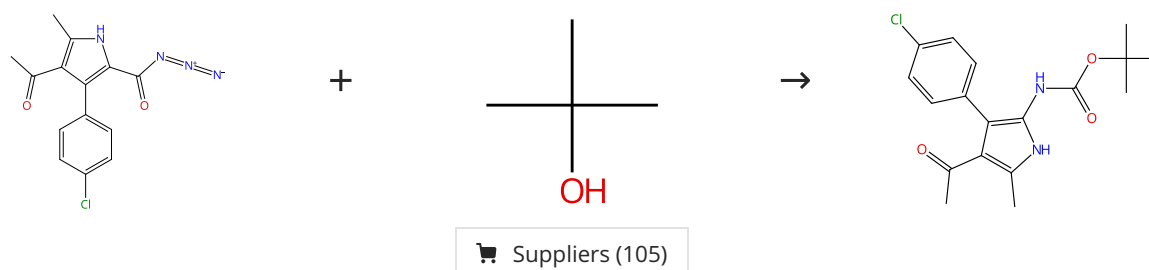
By: Bluechel, Christian; et al

Helvetica Chimica Acta (2003), 86(9), 2998-3036.

1.1 Solvents: Toluene; 2 h, 100 °C

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-21706392

Steps: 1 Yield: 100%

2H-Azirine-2-carbonyl Azides: Preparation and Use as N-Heterocyclic Building Blocks

By: Funt, Liya D.; et al

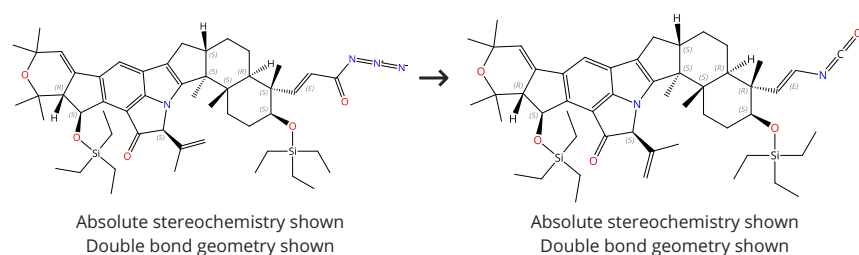
Journal of Organic Chemistry (2020), 85(6), 4182-4194.

1.1 Solvents: *tert*-Butanol; 1 h, reflux

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-807389

Steps: 1 Yield: 100%

Nodulisporic acid side-Chain modifications: access to the 2", 3", 4", and 6" registers

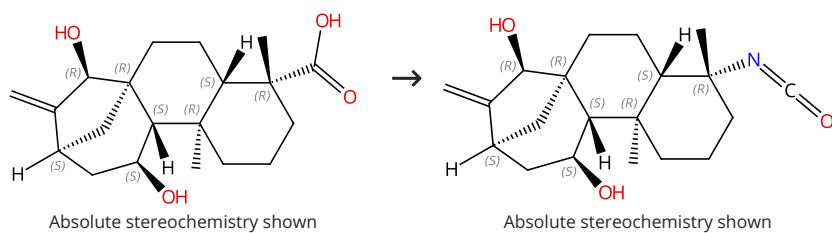
By: Chakravarty, Prasun K.; et al

Bioorganic & Medicinal Chemistry Letters (2003), 13(1), 147-150.

1.1 Reagents: Toluene; 80 °C

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (25)

31-449-CAS-21212389

Steps: 1 Yield: 100%

Kaurane diterpene derivative and its preparation method and medical application

1.1 **Reagents:** Triethylamine, Diphenylphosphoryl azide
Solvents: Anisole; rt; 1.5 h, rt → 90 °C

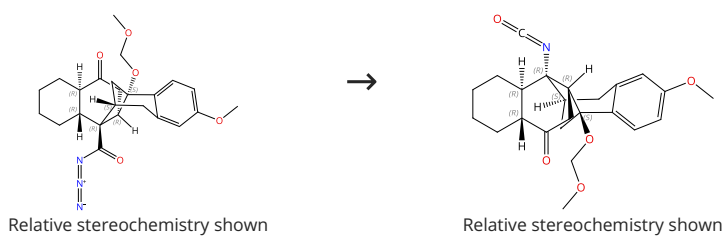
Assignee: Kunming Institute of Botany, Chinese Academy of Sciences

China, CN102603519 A 2012-07-25

PatentPak available

Scheme 17 (2 Reactions)

Steps: 1 Yield: 100%



31-449-CAS-15244270

Steps: 1 Yield: 100%

Approaches to the Synthesis of the Galbulimima Alkaloid Himandrine

1.1 **Solvents:** Toluene; 15 min, reflux

By: O'Connor, Patrick D.; et al

Australian Journal of Chemistry (2010), 63(10), 1477-1491.

31-449-CAS-9382611

Steps: 1 Yield: 100%

Synthesis of the Himandrine Skeleton

1.1 **Solvents:** Toluene; 15 min, reflux

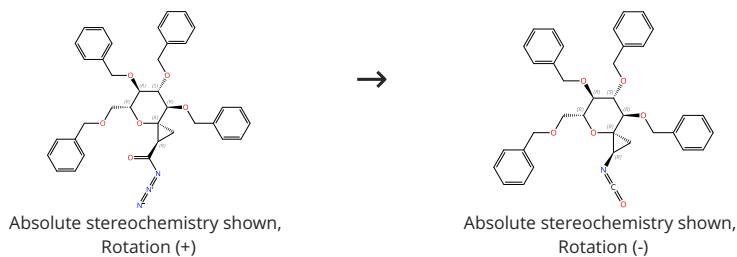
By: O'Connor, Patrick D.; et al

Experimental Protocols

Organic Letters (2004), 6(5), 703-706.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



31-449-CAS-13159686

Steps: 1 Yield: 100%

Synthesis of monosaccharide-derived spirocyclic cyclopropylamines and their evaluation as glycosidase inhibitors

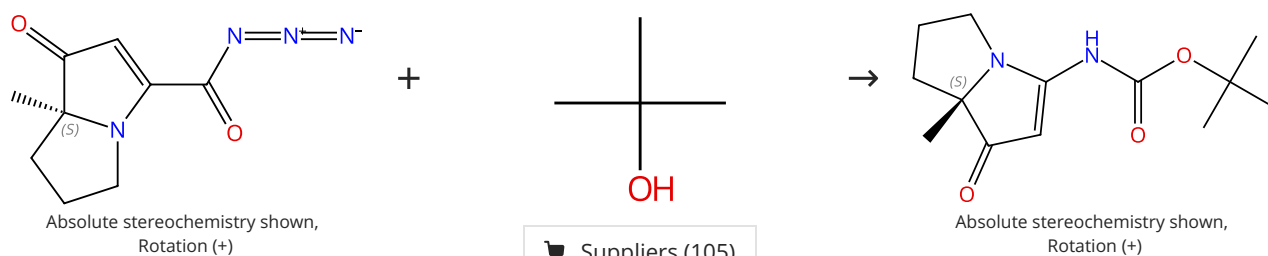
1.1 **Solvents:** Toluene; 2 h, 100 °C

By: Bluechel, Christian; et al

Helvetica Chimica Acta (2003), 86(9), 2998-3036.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (105)

31-449-CAS-8593396

Steps: 1 Yield: 99%

Synthesis of NP25302

1.1 Solvents: Toluene; 1 h, 80 °C; cooled

1.2 Reagents: Sodium bicarbonate
Solvents: Water; rt

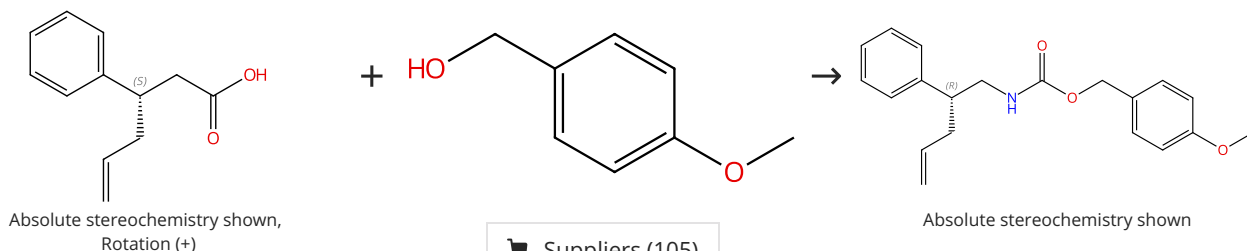
By: Stevens, Kiri; et al

Organic Letters (2011), 13(22), 5964-5967.

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (105)

31-304-CAS-18289935

Steps: 1 Yield: 99%

Synthesis of a macrocycle based on Linked Amino Acid
Mimetics (LAAM)1.1 Reagents: Triethylamine, Diphenylphosphoryl azide
Solvents: Toluene; overnight, reflux

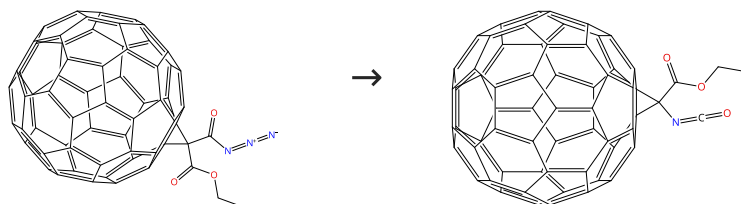
By: Maxwell, David S.; et al

Tetrahedron Letters (2013), 54(43), 5799-5801.

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



31-449-CAS-1035457

Steps: 1 Yield: 99%

Novel fullerene cyclopropanation reaction

1.1 Solvents: Toluene; 3 h, 80 °C

Assignee: Luna Innovations Incorporated

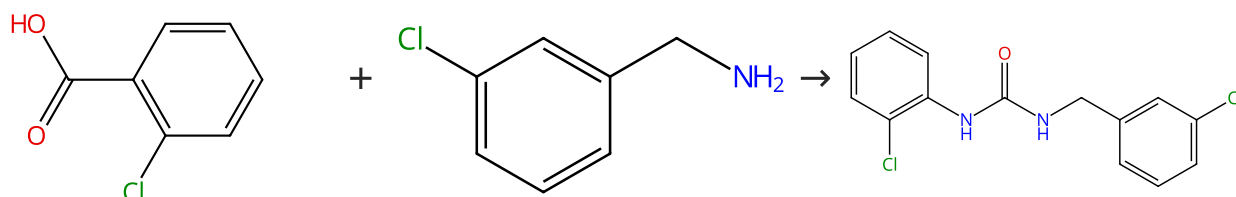
Experimental Protocols

World Intellectual Property Organization, WO2010014209 A2
2010-02-04

PatentPak available

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (81)

Suppliers (4)

31-316-CAS-12998575

Steps: 1 Yield: 99%

Thiol on silica as a 'catch and release' support for isocyanates to afford ureas

By: Bolshan, Yuri; et al

Tetrahedron Letters (2007), 48(28), 4925-4927.

1.1 Reagents: Triethylamine, Diphenylphosphoryl azide

Solvents: Toluene; 3 h, rt; 2 h, reflux

1.2 overnight, reflux

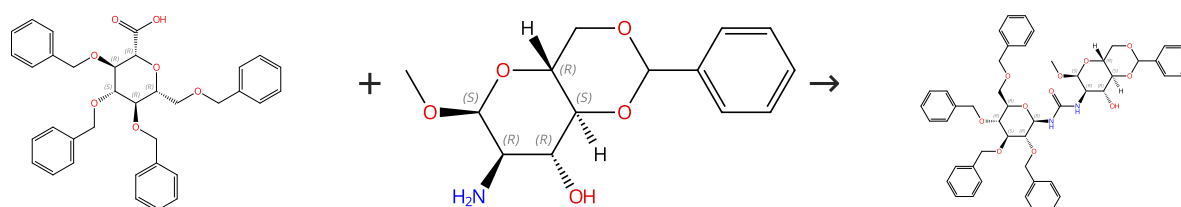
1.3 Reagents: Triethylamine

Solvents: Acetonitrile; overnight, 70 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (+)

Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (3)

31-316-CAS-18075431

Steps: 1 Yield: 99%

Novel synthesis of oligosaccharides linked with carbamate and urea bonds utilizing modified Curtius rearrangement

By: Sawada, Daisuke; et al

Tetrahedron (2008), 64(37), 8780-8788.

1.1 Reagents: Triethylamine, Diphenylphosphoryl azide

Solvents: Benzene; 1 h, reflux

1.2 Solvents: Acetonitrile; rt; 5 h, reflux; reflux → 0 °C

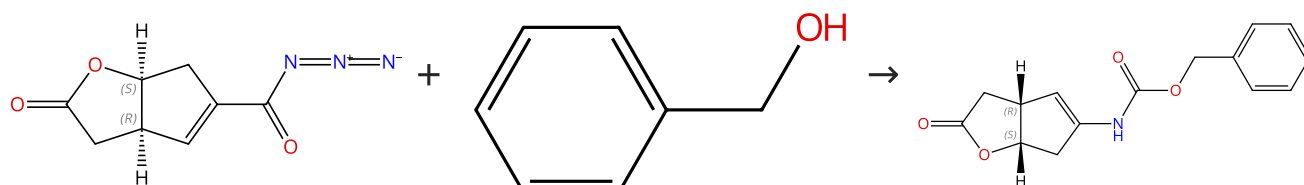
1.3 Reagents: Ammonium chloride

Solvents: Water; 0 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

Suppliers (149)

31-614-CAS-41497483

Steps: 1 Yield: 99%

Streamlined transformations of bicyclic enal intermediate - formal synthesis of Δ¹²-Prostaglandin J₃

By: Pelss, Andrejs; et al

Tetrahedron Letters (2024), 146, 155176.

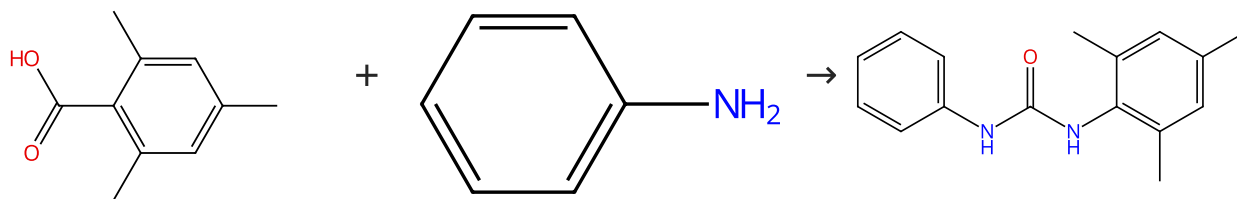
1.1 Solvents: Toluene; 1.5 h, rt → 80 °C; 80 °C → rt

1.2 Solvents: Toluene; 3 h, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (113)

Suppliers (7)

31-316-CAS-21055601

Steps: 1 Yield: 99%

Scalable, One-Pot, Microwave-Accelerated Tandem Synthesis of Unsymmetrical Urea Derivatives

1.1 **Reagents:** Triethylamine, Diphenylphosphoryl azide
Solvents: Toluene; 1 min, 100 °C

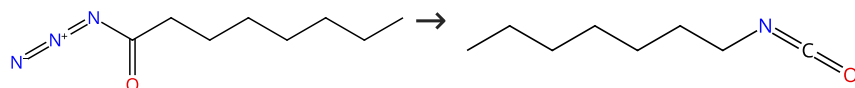
By: Kulkarni, Abhijit R.; et al

Experimental Protocols

Journal of Organic Chemistry (2017), 82(2), 992-999.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (48)

31-449-CAS-8885815

Steps: 1 Yield: 99%

Radical Azidonation of Aldehydes1.1 **Solvents:** Toluene

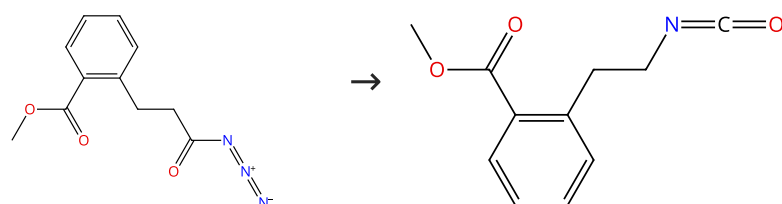
By: Marinescu, Lavinia; et al

Experimental Protocols

Journal of Organic Chemistry (2003), 68(24), 9453-9455.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Suppliers (8)

31-449-CAS-14208144

Steps: 1 Yield: 98%

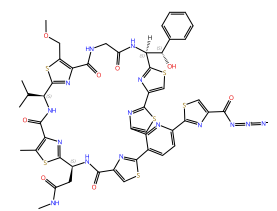
New synthetic methodology for construction of the 3,4-dihydroisoquinolinone skeleton: A key structure for isoquinoline alkaloids1.1 **Solvents:** Benzene; 1 h, reflux

By: Mujde, Berk; et al

Phytochemistry Letters (2011), 4(4), 407-410.

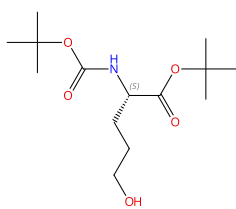
Scheme 28 (1 Reaction)

Steps: 1 Yield: 98%

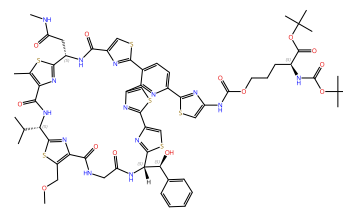


Absolute stereochemistry shown

+

Absolute stereochemistry shown,
Rotation (-)

→



Absolute stereochemistry shown

Suppliers (48)

31-449-CAS-4535678

Steps: 1 Yield: 98%

Antibacterial optimization of 4-aminothiazolyl analogues of the natural product GE2270 A: Identification of the cycloalkyl carboxylic acids

By: LaMarche, Matthew J.; et al

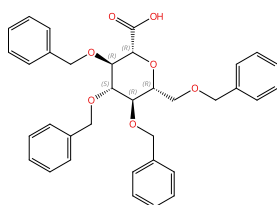
Journal of Medicinal Chemistry (2011), 54(23), 8099-8109.

1.1 Solvents: Toluene; 16 h, 80 °C

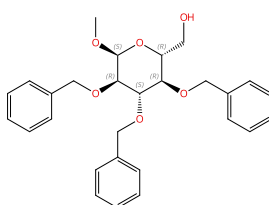
Experimental Protocols

Scheme 29 (2 Reactions)

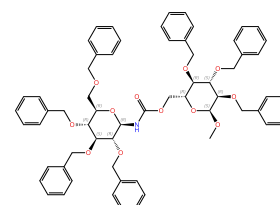
Steps: 1 Yield: 98%

Absolute stereochemistry shown,
Rotation (+)

+

Absolute stereochemistry shown,
Rotation (+)

→

Absolute stereochemistry shown,
Rotation (+)

Suppliers (64)

31-304-CAS-18152675

Steps: 1 Yield: 98%

Novel synthesis of oligosaccharides linked with carbamate and urea bonds utilizing modified Curtius rearrangement

By: Sawada, Daisuke; et al

Tetrahedron (2008), 64(37), 8780-8788.

1.1 Reagents: Silver carbonate, Potassium carbonate, Diphenyl phosphonyl azide
Solvents: Benzene; 8 h, reflux

1.2 Reagents: Ammonium chloride
Solvents: Water; 0 °C

Experimental Protocols

31-304-CAS-18228473

Steps: 1 Yield: 98%

Novel synthesis of carbamate-linked oligosaccharides by a modified curtius rearrangement

By: Sawada, Daisuke; et al

European Journal of Organic Chemistry (2007), (7), 1064-1068.

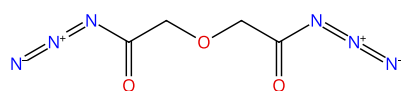
1.1 Reagents: Triethylamine, Diphenylphosphoryl azide
Solvents: Benzene; 8 h, reflux

1.2 Reagents: Ammonium chloride
Solvents: Water; 0 °C

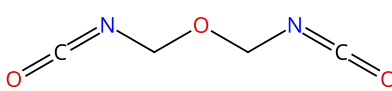
Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 98%



→



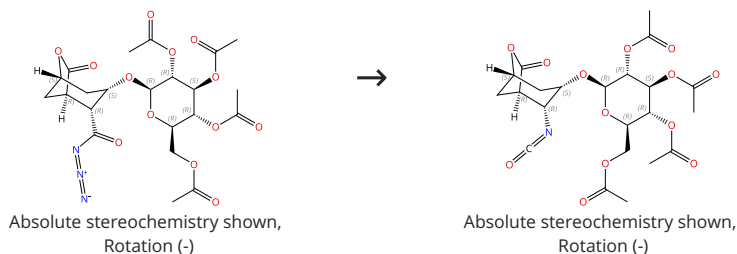
Supplier (1)

Suppliers (2)

31-449-CAS-9681888	Steps: 1 Yield: 98%	Diaminodimethyl ether
No Data Available		By: Popov, N. I.; et al Zhurnal Organicheskoi Khimii (1980), 16(7), 1551-2.

Scheme 31 (2 Reactions)

Steps: 1 Yield: 98%



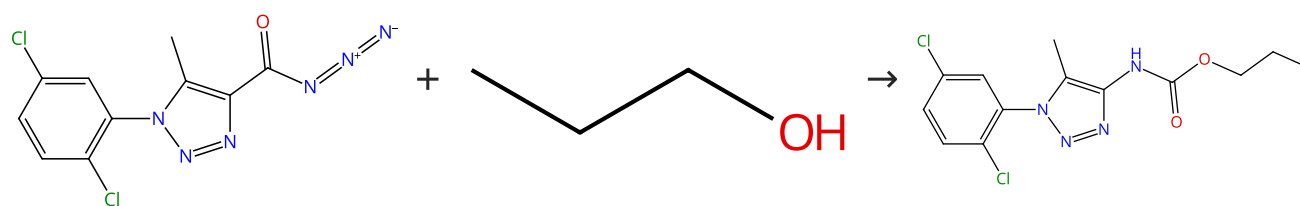
Supplier (1)

31-449-CAS-2474897	Steps: 1 Yield: 98%	Studies related to carba-pyranoses: a strategy for the synthesis of β-1,3-glycosidically linked aminomonocarba-disaccharides
1.1 Solvents: Benzene		By: Larsen, David S.; et al Journal of the Chemical Society, Perkin Transactions 1 (2001), (18), 2204-2212.

31-449-CAS-3465628	Steps: 1 Yield: 98%	The synthesis of pseudo-sugar disaccharides: a Diels-Alder approach to monocarba-disaccharides
1.1 Solvents: Benzene		By: Larsen, David S.; et al Tetrahedron Letters (1993), 34(50), 8151-4.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 98%



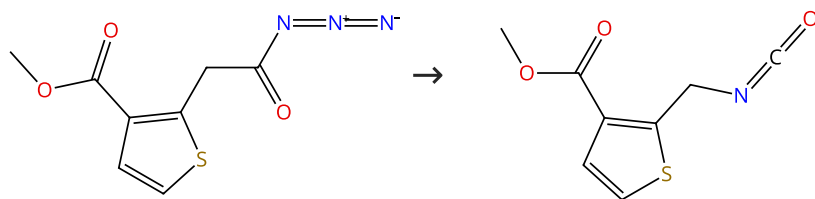
Suppliers (2)

Suppliers (155)

31-449-CAS-14928092	Steps: 1 Yield: 98%	The synthesis of some new N-[1-(2,5-dichlorophenyl)-5-methyl-1,2,3-triazol-4-yl]carbamic acid ester derivatives
1.1 80 - 100 °C		By: Dong, Heng-shan; et al Journal of the Chinese Chemical Society (Taipei, Taiwan) (2003), 50(6), 1209-1213.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

31-449-CAS-13862052

Steps: 1 Yield: 98%

A novel class of compounds: synthesis of 5,5'-carbonyl-bis(5,6-dihydro-4H-furo- and thieno-[2,3-c]pyrrol-4-ones)

1.1 Solvents: Benzene; 75 min, reflux

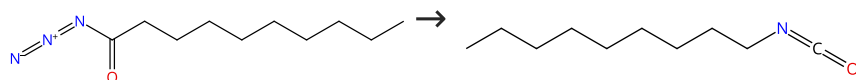
Experimental Protocols

By: Koza, Gani; et al

Tetrahedron (2011), 67(45), 8679-8684.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (4)

Suppliers (14)

31-449-CAS-4623476

Steps: 1 Yield: 98%

Radical Azidonation of Aldehydes

1.1 Solvents: Toluene

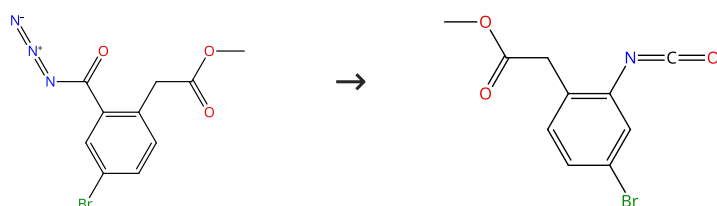
Experimental Protocols

By: Marinescu, Lavinia; et al

Journal of Organic Chemistry (2003), 68(24), 9453-9455.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (2)

31-449-CAS-10865225

Steps: 1 Yield: 98%

Synthesis of indolin-2-one, isoindolin-1-one, and indole derivatives from homophthalic acid

1.1 Solvents: Benzene; 1 h, reflux

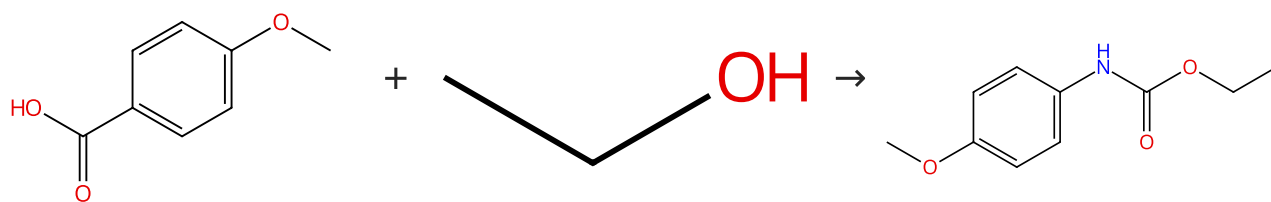
Experimental Protocols

By: Kilikli, Alper A.; et al

Synthesis (2011), (22), 3697-3705.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (111)

Suppliers (618)

Suppliers (16)

31-449-CAS-21298854

Steps: 1 Yield: 98%

Direct Conversion of Carboxylic Acids to Various Nitrogen-Containing Compounds in the One-Pot Exploiting Curtius Rearrangement

By: Kumar, Arun; et al

Journal of Organic Chemistry (2019), 84(17), 11323-11334.

1.1 **Reagents:** Diphenylphosphinic chloride, 1,8-Diazabicyclo [5.4.0]undec-7-ene

Solvents: Toluene; 5 min, 0 °C

1.2 **Reagents:** Sodium azide

Catalysts: 4-(Dimethylamino)pyridine

Solvents: Dimethylformamide; 0 °C

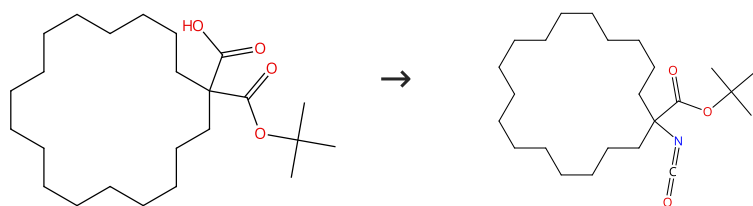
1.3 90 °C

1.4 **Reagents:** Hydrochloric acid

Solvents: Water; neutralized

Scheme 37 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Supplier (1)

31-449-CAS-18277877

Steps: 1 Yield: 98%

α,α -disubstituted glycines bearing a large hydrocarbon ring: peptide self-assembly through hydrophobic recognition

By: Ohwada, Tomohiko; et al

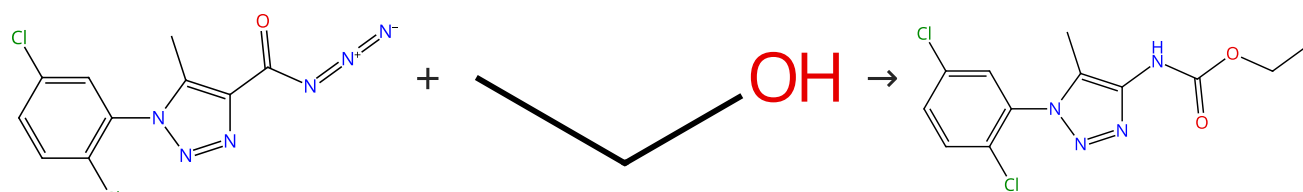
Chemistry - A European Journal (2004), 10(3), 617-626.

1.1 **Reagents:** Triethylamine, Diphenylphosphoryl azide

Solvents: Benzene; 3 h, reflux

Scheme 38 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (618)

31-449-CAS-5499711

Steps: 1 Yield: 97%

The synthesis of some new N-[1-(2,5-dichlorophenyl)-5-methyl-1,2,3-triazol-4-yl]carbamic acid ester derivatives

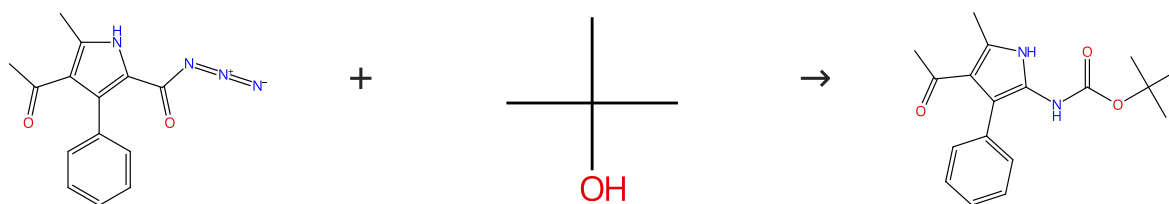
By: Dong, Heng-shan; et al

Journal of the Chinese Chemical Society (Taipei, Taiwan) (2003), 50(6), 1209-1213.

1.1 80 - 100 °C

Scheme 39 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (105)

31-449-CAS-21706390

Steps: 1 Yield: 97%

2H-Azirine-2-carbonyl Azides: Preparation and Use as N-Heterocyclic Building Blocks1.1 Solvents: *tert*-Butanol; 1 h, reflux

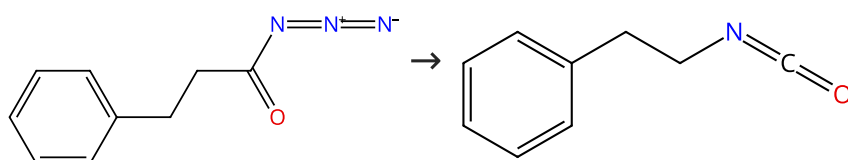
Experimental Protocols

By: Funt, Liya D.; et al

Journal of Organic Chemistry (2020), 85(6), 4182-4194.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (3)

Suppliers (57)

31-449-CAS-2501922

Steps: 1 Yield: 97%

Radical Azidonation of Aldehydes

1.1 Solvents: Toluene

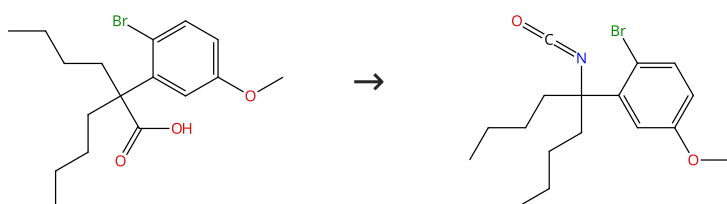
Experimental Protocols

By: Marinescu, Lavinia; et al

Journal of Organic Chemistry (2003), 68(24), 9453-9455.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 97%



31-449-CAS-18054593

Steps: 1 Yield: 97%

Synthesis of a mitochondria-targeted spin trap using a novel Parham-type cyclization1.1 Reagents: Triethylamine, Diphenylphosphoryl azide
Solvents: Toluene; 15 min, 0 °C; 0 °C → reflux; 18 h, reflux1.2 Reagents: Ammonium chloride
Solvents: Water

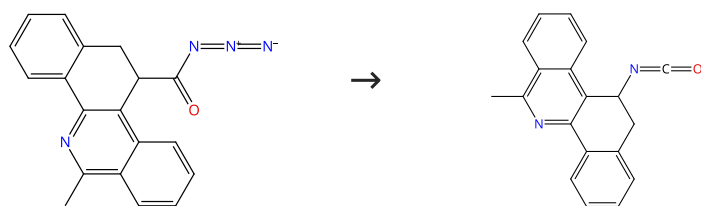
Experimental Protocols

By: Quin, Caroline; et al

Tetrahedron (2009), 65(39), 8154-8160.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

Suppliers (2)

31-449-CAS-13286144

Steps: 1 Yield: 96%

Benzophenanthridines. XI. Rodionov-Suvorov systems.
Synthesis of the first example of 11-aminobenzo[c]phenanthridines - 11-amino-6-methyl-11,12-dihydrobenzo[c]phenanthridine

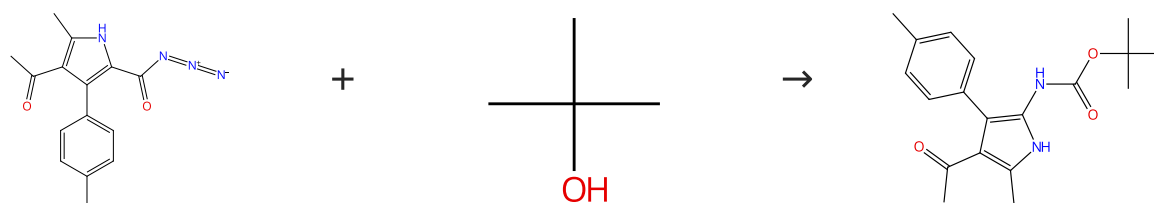
By: Maslennikova, L. V.; et al

Zhurnal Organicheskoi Khimii (1990), 26(2), 441-5.

1.1 Solvents: Benzene

Scheme 43 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

Suppliers (105)

31-449-CAS-21706395

Steps: 1 Yield: 96%

2H-Azirine-2-carbonyl Azides: Preparation and Use as N-Heterocyclic Building Blocks

By: Funt, Liya D.; et al

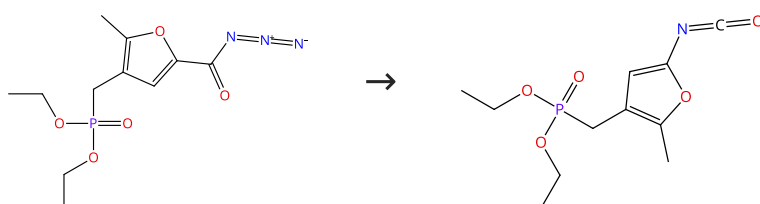
Journal of Organic Chemistry (2020), 85(6), 4182-4194.

1.1 Solvents: *tert*-Butanol; 1 h, reflux

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

31-449-CAS-7455558

Steps: 1 Yield: 96%

Diethoxyphosphorylmethylfurans in Curtius reaction

By: Pevzner, L. M.

Russian Journal of General Chemistry (2011), 81(1), 56-70.

1.1 Solvents: Toluene; 1 h, 105 - 110 °C

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 96%



31-449-CAS-1204098

Steps: 1 Yield: 96%

Synthesis of 1,3,5,7-tetrasubstituted adamantanes

No Data Available

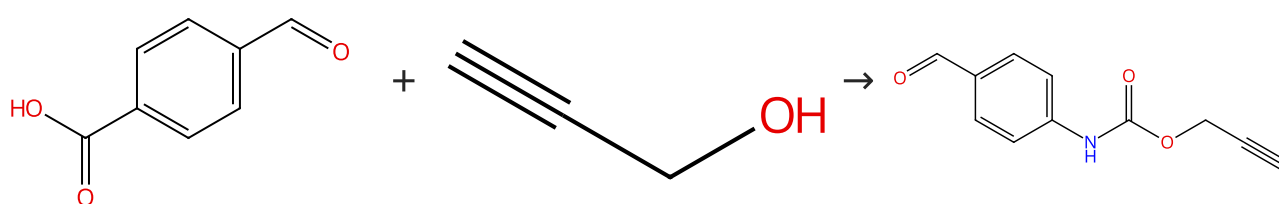
By: Moiseev, I. K.; et al

Experimental Protocols

Russian Journal of General Chemistry (Translation of Zhurnal Obshchei Khimii) (1999), 69(4), 673-674.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (101)

Suppliers (44)

Suppliers (4)

31-449-CAS-18151900

Steps: 1 Yield: 96%

Propylphosphonic anhydride (T3P)-mediated one-pot rearrangement of carboxylic acids to carbamates

1.1 **Reagents:** Triethylamine, Azidotrimethylsilane, Propylphosphonic anhydride
Solvents: Tetrahydrofuran; 2 - 3 h, reflux

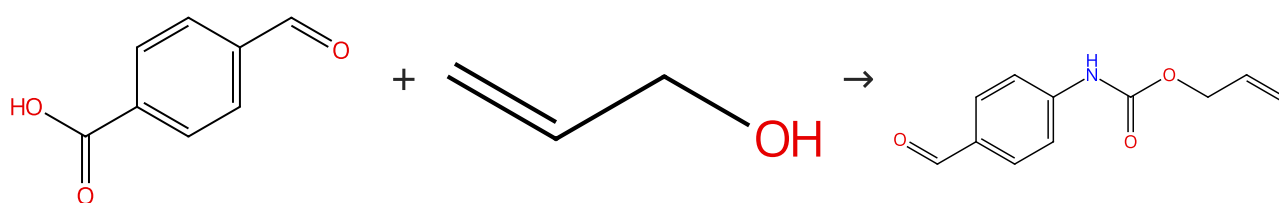
By: Augustine, John Kallikat; et al

Synthesis (2011), (9), 1477-1483.

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (101)

Suppliers (47)

Suppliers (5)

31-449-CAS-18130944

Steps: 1 Yield: 96%

Propylphosphonic anhydride (T3P)-mediated one-pot rearrangement of carboxylic acids to carbamates

1.1 **Reagents:** Triethylamine, Azidotrimethylsilane, Propylphosphonic anhydride
Solvents: Tetrahydrofuran; 2 - 3 h, reflux

By: Augustine, John Kallikat; et al

Synthesis (2011), (9), 1477-1483.

Experimental Protocols